

Differential Diagnosis of Atopic Dermatitis

Most Likely

- Contact dermatitis (allergic and irritant)
- Seborrheic dermatitis
- Scabies
- Psoriasis
- Ichthyosis vulgaris
- Keratosis pilaris
- Dermatophytosis

Consider

- Asteatotic eczema
- Lichen simplex chronicus
- Nummular dermatitis
- Juvenile palmar-plantar dermatosis
- Impetigo
- Drug eruptions
- Perioral dermatitis
- Pityriasis alba
- Photosensitivity disorders (hydroa vacciniforme, polymorphous light eruption, porphyrias)
- Molluscum dermatitis

Less Common/Rare Disorders Predominantly in Adolescents and Adults

- Cutaneous T-cell lymphoma (mycosis fungoides or Sézary syndrome)
- Human immunodeficiency virus–associated dermatoses
- Lupus erythematosus
- Dermatomyositis
- Graft-versus-host disease
- Pemphigus foliaceus
- Dermatitis herpetiformis
- Photosensitivity disorders (hydroa vacciniforme, polymorphous light eruption, porphyrias)

Complications

Ocular Problems

Eye complications associated with severe atopic dermatitis can lead to significant morbidity. Eyelid dermatitis and chronic blepharitis are commonly seen and may result in visual impairment due to corneal scarring. Atopic keratoconjunctivitis is typically bilateral and can cause disabling symptoms such as itching, burning, tearing, and copious mucoid discharge.

Infants presenting in the first year of life with failure to thrive, diarrhea, generalized scaling erythematous rash, and recurrent cutaneous or systemic infections should be evaluated for severe combined immunodeficiency syndrome. Wiskott-Aldrich syndrome, an X-linked recessive disorder, presents with cutaneous findings nearly indistinguishable from atopic dermatitis. It is associated with thrombocytopenia, variable abnormalities in humoral and cellular immunity, and recurrent severe bacterial infections.

Hyper-IgE syndrome is characterized by elevated serum IgE levels, defective T-cell function, and recurrent deep-seated bacterial infections, including cutaneous abscesses caused by *Staphylococcus aureus* or pruritic skin disease due to *S. aureus* pustulosis, or recalcitrant dermatophytosis. A papulopustular eruption

on the face and scalp may appear in early life. Although *S. aureus* is a major pathogen, infections with other bacteria, viruses, and fungi may occur—especially in patients on long-term anti-staphylococcal antibiotic prophylaxis. Additional features include pneumonia with pneumatocele formation, dental anomalies with retained primary teeth, bone fractures, and osteopenia.

It is important to recognize that adults presenting with eczematous dermatitis without a history of childhood eczema, respiratory allergy, or atopic family history may have allergic contact dermatitis. A contact allergen should be suspected in any patient whose dermatitis fails to respond to appropriate therapy. Notably, contact allergy to topical glucocorticoids and topical calcineurin inhibitors has been reported in patients with chronic dermatitis. Additionally, cutaneous T-cell lymphoma must be ruled out in adults with chronic, treatment-resistant dermatitis. Skin biopsies should ideally be obtained from three separate sites, as histology may show spongiosis and inflammatory infiltrate resembling atopic dermatitis.

Eczematous dermatitis has also been reported in association with human immunodeficiency virus infection and various infestations, such as scabies. Other conditions that may mimic atopic dermatitis include psoriasis, ichthyoses, and seborrheic dermatitis.